



Original Investigation | Oncology

Survival and Recurrence With GLP-1 Receptor Agonists in Breast Cancer

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Abstract

IMPORTANCE Patients with breast cancer (BC) with comorbid obesity or type 2 diabetes (T2D) experience poorer survival. Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) are approved to treat these comorbidities; however, their associations with BC survival and recurrence remain unclear.

OBJECTIVE To evaluate the association between GLP-1 RA use and 10-year all-cause mortality and recurrence-free survival (RFS) over the 10-year follow-up period, as well as 5- and 10-year all-cause mortality and RFS probabilities among patients with BC.

DESIGN, SETTING, AND PARTICIPANTS This retrospective cohort study used TriNetX US Collaborative Network data from women (≥ 18 years) with BC from 68 health care organizations who received a diagnosis between April 1, 2006, and April 1, 2023. Propensity score matching balanced characteristics. Data were analyzed between September 16 and October 3, 2025.

EXPOSURES GLP-1 RA use (≥ 2 prescriptions) during the 6 months before and any time after the index diagnosis; nonuse (0 entries).

MAIN OUTCOMES AND MEASURES The primary outcome was all-cause mortality, and the secondary outcome was RFS. Cox proportional hazards regression model–estimated hazard ratios (HRs) were restricted to 10 years. Kaplan-Meier estimators were used to calculate 5- and 10-year all-cause mortality and RFS probabilities. Prespecified subgroup (postmenopausal) and landmark (6- and 12-month) analyses were conducted.

RESULTS The study comprised 841 831 eligible patients with BC (mean [SD] age, 69.1 [12.2] years). After exclusions and 1:1 propensity score matching, 3 cohorts were identified: 1610 patients for GLP-1 RA use vs nonuse (patients with obesity [body mass index ≥ 30]), 2323 patients for GLP-1 RA use vs insulin or metformin (patients with T2D), and 4052 patients for GLP-1 RA use vs sodium-glucose cotransporter 2 inhibitors (patients with T2D). Among patients with obesity, GLP-1 RAs were associated with lower hazard of all-cause mortality (HR, 0.35; 95% CI, 0.21-0.58; $P < .001$) and RFS (HR, 0.44; 95% CI, 0.30-0.64; $P < .001$) over a 10-year follow-up period. Among patients with T2D, GLP-1 RAs vs insulin or metformin were associated with lower hazard of all-cause mortality (HR, 0.09; 95% CI, 0.06-0.15; $P < .001$) and RFS (HR, 0.33; 95% CI, 0.21-0.50; $P < .001$). No significant differences were observed between GLP-1 RA and sodium-glucose cotransporter 2 inhibitor groups. Subgroup and landmark analyses yielded similar findings.

CONCLUSIONS AND RELEVANCE In this cohort study of patients with BC, findings suggested a potential association between GLP-1 RA use and improved outcomes among patients with BC who have obesity and related metabolic conditions. These findings support further evaluation of GLP-1 RA therapy in randomized clinical trials.

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Key Points

Question Is there a potential benefit associated with the use of glucagon-like peptide-1 receptor agonists (GLP-1 RAs) in female breast cancer survival and cancer recurrence among patients with obesity or with type 2 diabetes (T2D)?

Findings In this cohort study using propensity score–matched data from 841 831 patients with breast cancer, GLP-1 RA use vs nonuse was significantly associated with improved survival and reduced recurrence among patients with obesity. Similar benefits were seen among patients with T2D when compared with other antidiabetic medications.

Meaning This study suggests that GLP-1 RAs may offer protective benefits beyond glycemic and weight control, potentially improving survival and recurrence risk in some female patients with breast cancer.

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Introduction

Breast cancer (BC) is the most prevalent cancer among women in the US, accounting for more than 30% of all cancer cases. Preexisting conditions such as obesity and type 2 diabetes (T2D) increase the risk of poor prognosis, disease progression, cancer recurrence, and reduced survival.¹⁻¹⁰ Although understudied, interventions for weight reduction to address obesity among patients with BC may improve outcomes.¹¹⁻¹⁴ Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) are established pharmacologic treatments for T2D and weight loss.¹⁵⁻¹⁷ Their potential protective effects on BC survival and recurrence remain unclear.

Obesity can lead to insulin resistance that results in T2D, further affecting BC outcomes. Previous studies have shown the benefit of GLP-1 RAs in reducing obesity¹⁸⁻²³ and related comorbidities²⁴ (eg, cardiovascular events independent of weight loss),^{25,26} yet few studies have focused on all-cause mortality and recurrence-free survival (RFS) outcomes in cohorts of patients with cancer.¹⁴ Preclinical studies demonstrate the anticancer benefits of GLP-1 RAs inhibiting tumor growth in certain malignant neoplasms, including BC.^{27,28} In addition, weight loss treatment and weight loss surgery after BC diagnosis are associated with better cardiovascular outcomes and increased survival, respectively.^{11,12}

The effect of GLP-1 RAs on BC outcomes exclusively has yet to be reported in the literature, to our knowledge. To address this gap, this study uses electronic health record (EHR) data across multiple US health care organizations from adult female patients with BC to investigate the extent to which GLP-1 RA use is associated with decreased all-cause mortality and improved RFS (or metastasis).

Methods

This retrospective, comparative cohort study was exempt from review and informed consent was waived by Virginia Commonwealth University institutional review board ethics review, as this study used only aggregated, anonymized data with a deidentification process formally attested by a qualified expert, as defined in §164.514(b)(1) of the Health Insurance Portability and Accountability Act Privacy Rule.²⁹ This study was conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for cohort studies.

Data Source

We used data from TriNetX LLC,^{30,31} a multicenter federated health research network. This network aggregates deidentified data from EHRs, providing access to patient demographic characteristics, procedures, medications, laboratory test values, and diagnoses from more than 174 million patients and across more than 86 health care organizations in the US, including hospitals, outpatient centers, and primary care clinics.³¹ Retrospective cohorts were formed and defined using the *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision, Clinical Modification* (ICD-10-CM), Anatomical Therapeutic Chemical classification, and *Current Procedural Terminology* codes and RxNorm in the TriNetX cohort builder. Cohort definitions and query codes are shown in eTable 1 in Supplement 1. Analysis was performed using TriNetX built-in analytic tools. Demographic data on self-reported race and ethnicity categories in the EHRs (American Indian or Alaska Native, Asian, Black or African American, Hispanic or Latino, Native Hawaiian or Other Pacific Islander, White, other [defined as multiracial and any race not otherwise specified], or unknown) were included in accordance with the Office of Management and Budget standards.³²

Study Population

Data collection and analysis occurred between September 16 and October 3, 2025. At the time of analysis, the TriNetX US Collaborative Network included 118 970 170 patients across 68 health care organizations. Applying the inclusion criteria for most recent diagnosis of stages I to III BC (ICD-10-CM code C50) and excluding patients with any instance of stage IV BC, metastatic disease, or other prior

malignant neoplasms resulted in a study population of 841 831 patients (eFigures 1 and 2 in [Supplement 1](#)). Patients who received a BC diagnosis between April 1, 2006, and April 1, 2023 (index date), were included to align with the timeframe of 1 year after the US Food and Drug Administration (FDA) approval of GLP-1 RAs for diabetes management in 2005 and for chronic weight management in adults in 2021,^{15,17} similar to previously described methods.^{33,34} The index date for all groups was the date of BC diagnosis (documented medical visit). GLP-1 RA use was defined as an exposure of 2 or more prescription instances, within 6 months before or any time after the index date; nonuse was defined as no GLP-1 RA prescriptions.

The study population was divided into 2 separate cohorts (eFigures 1 and 2 in [Supplement 1](#)). The obesity cohort compared GLP-1 RA use vs nonuse or other weight-loss medications among patients with obesity (body mass index [BMI] ≥ 30 ; calculated as weight in kilograms divided by height in meters squared) and excluding individuals with T2D or a recent hemoglobin HbA_{1c} (HbA_{1c}) greater than 6.5% (to convert to proportion of total hemoglobin, multiply by 0.01). The T2D cohort compared GLP-1 RA use vs insulin or metformin and GLP-1 RA use vs sodium-glucose cotransporter 2 (SGLT2) inhibitors among patients with T2D or an HbA_{1c} of 6.5% or more, excluding underweight patients (BMI ≤ 18).

Study Outcomes

The primary outcome was all-cause mortality, defined as death from any cause, assessed from the EHRs and the TriNetX mortality linkage. The secondary outcome was RFS, defined as the time from the index date to the first occurrence of metastatic (distant) recurrence, identified using *ICD-10-CM* codes C77 to C79.^{35,36} The all-cause mortality and RFS outcome definitions and corresponding codes are detailed in eTable 1 in [Supplement 1](#). Outcomes were assessed over a 10-year follow-up period.

Covariates

We used propensity score matching (PSM) to balance cohorts on covariates (demographic characteristics, diagnosis, procedures, and medications) that were selected based on literature review, evidence-based practices, and data availability.³⁷⁻³⁹ The same covariates were used for all groups, including subgroups (eTable 1 in [Supplement 1](#)). The standardized codes and characteristics of the additional comparison groups before and after matching are presented in eTables 1 to 7 in [Supplement 1](#).

Statistical Analysis

Statistical analysis was conducted between September 16 and October 3, 2025. PSM was performed separately for each comparison group using 1:1 greedy nearest-neighbor matching with a caliper of 0.1, based on the covariates described. Before matching, differences were observed across cohorts for age at index, race and ethnicity distribution, and prevalence of obesity. After matching, all cohorts were well balanced, with standardized mean differences less than 0.1.

Survival analyses were conducted using Kaplan-Meier methods to derive unadjusted survival curves and hazard ratios (HRs). The statistical significance of Kaplan-Meier survival curve differences was calculated using the log-rank test (2-sided $P < .05$). The Cox proportional hazards regression models estimated adjusted HRs to assess comparisons between cohort groups after adjusting for covariates. All measures were reported with 95% CIs. Follow-up for Cox proportional hazards regression models were restricted to 10 years. Patients without an event by 10 years were administratively censored. Kaplan-Meier estimates were used to calculate 5- and 10-year all-cause mortality and RFS probabilities. Because TriNetX reports the IQR as a single value, the 25th and 75th percentiles were approximated from the median and IQR assuming a symmetric distribution. Data within TriNetX are harmonized and standardized across participating health systems, with analyses conducted on available data without imputation. The potential for residual confounding from unmeasured variables cannot be excluded. Subgroup analyses were conducted for postmenopausal patients with BC. To address potential immortal-time bias, landmark analyses, defined as a

documented medical follow-up visit that occurred at least 6 or 12 months after any instance of GLP-1 RA use, were performed. All models accounted for censoring at the time of disenrollment or loss to follow-up.

Results

Population Characteristics

Of the 841 831 patients (mean [SD] age, 69.1 [12.2] years; 2610 American Indian or Alaska Native [0.3%], 37 461 Asian [4.5%], 101 946 Black or African American [12.1%], 47 058 Hispanic or Latino [5.6%], 6314 Native Hawaiian or Other Pacific Islander [0.8%], 603 929 White [71.7%]; 62 801 unknown race or ethnicity [7.5%]; and 26 770 other race or ethnicity [3.2%]) with a diagnosis of BC, 573 443 eligible patients with obesity and 652 001 patients with T2D were included in PSM (**Table 1**). Baseline characteristics for PSM were the same for all-cause mortality and RFS outcomes (eTables 2-7 in [Supplement 1](#)). The median follow-up times after PSM were comparable across cohort comparisons groups (eTable 8 in [Supplement 1](#)). The PSM outcomes for RFS, postmenopausal subgroup, and landmark analyses are shown in **Table 2**. The subgroup results are descriptive and should be interpreted cautiously.

Comparison of GLP-1 RA Use vs Nonuse Among Patients With BC and Obesity

Prior to PSM, the GLP-1 RA use group included 4843 patients, and the nonuse group included 7473 patients. After PSM, 1610 patients were included in each group (**Table 1**), with comparable median follow-up times of 819 days (IQR, 489.5-1148.5 days) for GLP-1 RA use and 931 days (IQR, 515.0-1347.0 days) for nonuse (eTable 8 in [Supplement 1](#)). The 5- and 10-year survival probabilities were 97.4% and 96.0% for GLP-1 RA use, respectively, and 93.2% and 88.6% for nonuse, respectively (**Figure 1A**). In unadjusted analyses, GLP-1 RA use was associated with lower risk of all-cause mortality (HR, 0.35; 95% CI, 0.21-0.58; $P < .001$) over the 10-year follow-up period (**Figure 2**). In adjusted models, GLP-1 RA use remained associated with a significantly lower hazard of all-cause mortality compared with nonuse (HR, 0.42; 95% CI, 0.28-0.62; $P < .001$) (eTable 9 in [Supplement 1](#)).

For RFS, prior to PSM, there were 4846 patients in the GLP-1 RA use group and 7473 patients in the nonuse group (eTable 5 in [Supplement 1](#)). After PSM, groups were balanced with 1568 patients in each. Survival probabilities at 5 and 10 years were 96.0% and 87.7%, respectively, for GLP-1 RA use and 91.3% and 90.6%, respectively, for nonuse (**Figure 1D**). In unadjusted analyses, GLP-1 RA use was associated with a statistically significantly lower hazard of recurrence (HR, 0.44; 95% CI, 0.30-0.64; $P < .001$) (**Figure 3**). However, the number of patients remaining at risk after 5 years was low. Specifically, 167 GLP-1 RA users and 160 nonusers remained at risk at day 1762, as most patients were administratively censored at year 5. In the adjusted model, GLP-1 RA use vs nonuse remained significant (HR, 0.56; 95% CI, 0.46-0.67; $P < .001$) (eTable 10 in [Supplement 1](#)).

Comparison of GLP-1 RA vs Insulin or Metformin Among Patients With BC and T2D

Prior to PSM, the GLP-1 RA group included 2383 patients, and the insulin or metformin group included 51 397 patients. After PSM, groups were balanced with 2323 patients in each group (**Table 1**). Survival probabilities for 5 and 10 years were both 96.9% for GLP-1 RA and 82.3% and 76.4%, respectively, for insulin or metformin (**Figure 1B**). The 10-year survival probability for the GLP-1 RA group remained 96.9% because participants were administratively censored. GLP-1 RA use was associated with a lower hazard of all-cause mortality in unadjusted analyses (HR, 0.09; 95% CI, 0.06-0.15; $P < .001$) (**Figure 2**). In the adjusted Cox proportional hazards regression model, GLP-1 RA use maintained this protective association for all-cause mortality (HR, 0.12; 95% CI, 0.08-0.18; $P < .001$) (eTable 9 in [Supplement 1](#)).

For RFS, prior to PSM, the GLP-1 RA group included 2383 patients, and the metformin or insulin group included 51 397 patients. After matching, groups were balanced with 2312 patients in each group (**Table 1**). Survival probability at both 5 and 10 years was 97.4% for GLP-1 RA use and 93.4% for

Table 1. Patient Characteristics Across Comparison Groups After Propensity Score Matching^a

Characteristic	GLP-1 RA use vs nonuse			GLP-1 RA vs insulin or metformin			GLP-1 RA vs SGLT2 inhibitors		
	No. (%)	No. (%)	SMD ^b	No. (%)	No. (%)	SMD ^b	No. (%)	No. (%)	SMD ^b
	GLP-1 RA use (n = 1610)	Nonuse (n = 1610)		GLP-1 RA (n = 2323)	Insulin or metformin (n = 2323)		GLP-1 RA (n = 4052)	SGLT2 inhibitors (n = 4052)	
Age at index, mean (SD), y	57.7 (10.7)	58.6 (11.3)	0.078	63.7 (9.8)	63.6 (11.4)	0.005	68.6 (9.1)	68.5 (9.5)	0.005
Race									
American Indian or Alaska Native	10 (0.6)	10 (0.6)	0.001	10 (0.4)	10 (0.4)	0.001	19 (0.5)	12 (0.3)	0.028
Asian	24 (1.5)	20 (1.2)	0.021	102 (4.4)	82 (3.5)	0.044	288 (7.1)	282 (7.0)	0.006
Black or African American	235 (14.6)	242 (15.0)	0.012	473 (20.4)	459 (19.8)	0.015	945 (23.3)	938 (23.1)	0.004
Native Hawaiian or Other Pacific Islander	10 (0.6)	10 (0.6)	0.001	22 (0.9)	28 (1.2)	0.025	47 (1.2)	39 (1.0)	0.019
White	1199 (74.5)	1188 (73.8)	0.016	1591 (68.5)	1625 (70.0)	0.032	2390 (59.0)	2412 (59.5)	0.011
Other ^c	60 (3.7)	59 (3.7)	0.003	49 (2.1)	45 (1.9)	0.012	149 (3.7)	145 (3.6)	0.005
Unknown race	84 (5.2)	94 (5.8)	0.027	78 (3.4)	76 (3.3)	0.005	214 (5.3)	224 (5.5)	0.011
Ethnicity									
Hispanic or Latino	101 (6.3)	100 (6.2)	0.003	128 (5.5)	115 (5.0)	0.025	365 (9.0)	356 (8.8)	0.008
Not Hispanic or Latino	1181 (73.4)	1203 (74.7)	0.031	1687 (72.6)	1714 (73.8)	0.026	3056 (75.4)	3038 (75.0)	0.01
Unknown ethnicity	328 (20.4)	307 (19.1)	0.033	508 (21.9)	494 (21.3)	0.015	631 (15.6)	658 (16.2)	0.018
Socioeconomic risks ^d	23 (1.4)	19 (1.2)	0.022	45 (1.9)	37 (1.6)	0.026	75 (1.9)	79 (1.9)	0.007
Lifestyle factors									
Nicotine dependence	55 (3.4)	46 (2.9)	0.032	94 (4.0)	110 (4.7)	0.034	241 (5.9)	228 (5.6)	0.014
Alcohol-related disorders	19 (1.2)	20 (1.2)	0.006	18 (0.8)	19 (0.8)	0.005	33 (0.8)	38 (0.9)	0.013
Breast cancer characteristics									
Carcinoma in situ of breast	119 (7.4)	109 (6.8)	0.024	203 (8.7)	220 (9.5)	0.025	330 (8.1)	346 (8.5)	0.014
ER-positive status	229 (14.2)	221 (13.7)	0.014	495 (21.3)	467 (20.1)	0.03	768 (19.0)	736 (18.2)	0.020
ER-negative status	57 (3.5)	53 (3.3)	0.014	79 (3.4)	71 (3.1)	0.019	135 (3.3)	144 (3.6)	0.012
Comorbidities									
Overweight and obesity ^e	560 (34.8)	287 (17.8)	0.392	997 (42.9)	607 (26.1)	0.359	1592 (39.3)	1034 (25.5)	0.297
Bariatric surgery	28 (1.7)	29 (1.8)	0.005	73 (3.1)	60 (2.6)	0.034	40 (1.0)	37 (0.9)	0.008
CKD	51 (3.2)	67 (4.2)	0.053	236 (10.2)	244 (10.5)	0.011	816 (20.1)	798 (19.7)	0.011
COPD	51 (3.2)	47 (2.9)	0.014	103 (4.4)	93 (4.0)	0.021	329 (8.1)	325 (8.0)	0.004
Cerebrovascular disease	43 (2.7)	45 (2.8)	0.008	81 (3.5)	64 (2.8)	0.042	331 (8.2)	311 (7.7)	0.018
Ischemic heart disease	72 (4.5)	72 (4.5)	0.001	202 (8.7)	201 (8.7)	0.002	765 (18.9)	763 (18.8)	0.001
Heart failure	44 (2.7)	36 (2.2)	0.032	136 (5.9)	135 (5.8)	0.002	653 (16.1)	643 (15.9)	0.007
Diseases of liver	83 (5.2)	88 (5.5)	0.014	241 (10.4)	233 (10.0)	0.011	360 (8.9)	355 (8.8)	0.004
Hypertensive diseases	507 (31.5)	514 (31.9)	0.009	1358 (58.5)	1333 (57.4)	0.022	2897 (71.5)	2885 (71.2)	0.007
Dyslipidemia	447 (27.8)	462 (28.7)	0.021	1227 (52.8)	1206 (51.9)	0.018	2624 (64.8)	2597 (64.1)	0.014
OSA	151 (9.4)	147 (9.1)	0.009	415 (17.9)	391 (16.8)	0.027	572 (14.1)	560 (13.8)	0.009
Medications									
Metformin	103 (6.4)	97 (6.0)	0.015	26 (1.1)	28 (1.2)	0.008	1687 (41.6)	1741 (43.0)	0.027
SGLT2 inhibitors	13 (0.8)	11 (0.7)	0.014	183 (7.9)	174 (7.5)	0.015	39 (1.0)	52 (1.3)	0.030

(continued)

Table 1. Patient Characteristics Across Comparison Groups After Propensity Score Matching^a (continued)

Characteristic	GLP-1 RA use vs nonuse			GLP-1 RA vs insulin or metformin			GLP-1 RA vs SGLT2 inhibitors		
	No. (%)			No. (%)			No. (%)		
	GLP-1 RA use (n = 1610)	Nonuse (n = 1610)	SMD ^b	GLP-1 RA (n = 2323)	Insulin or metformin (n = 2323)	SMD ^b	GLP-1 RA (n = 4052)	SGLT2 inhibitors (n = 4052)	SMD ^b
Treatment ^f									
Mastectomy procedures	66 (4.1)	59 (3.7)	0.023	85 (3.7)	80 (3.4)	0.012	148 (3.7)	152 (3.8)	0.005
Mastectomy, partial ^g	42 (2.6)	35 (2.2)	0.028	65 (2.8)	64 (2.8)	0.003	92 (2.3)	99 (2.4)	0.011
Radiotherapy	42 (2.6)	40 (2.5)	0.008	65 (2.8)	62 (2.7)	0.008	113 (2.8)	110 (2.7)	0.005
Chemotherapy	359 (22.3)	357 (22.2)	0.003	616 (26.5)	591 (25.4)	0.025	1141 (28.2)	1172 (28.9)	0.017
Aromatase inhibitors	243 (15.1)	235 (14.6)	0.014	441 (19.0)	426 (18.3)	0.017	797 (19.7)	813 (20.1)	0.010
Tamoxifen	95 (5.9)	103 (6.4)	0.021	105 (4.5)	104 (4.5)	0.002	141 (3.5)	131 (3.2)	0.014

Abbreviations: CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; ER, estrogen receptor; GLP-1 RA, glucagon-like peptide-1 receptor agonist; OSA, obstructive sleep apnea; SGLT2, sodium-glucose cotransporter 2; SMD, standardized mean difference.

^a Propensity score matching for demographics, diagnosis, procedure, and medication.

^b SMD less than 0.10 indicates that the 2 comparison groups were well balanced.

^c Other race was defined as multiracial and any race not otherwise specified.

^d Socioeconomic risks refer to persons with potential health hazards related to socioeconomic and psychosocial circumstances (*International Statistical Classification of Diseases and Related Health Problems, Tenth Revision, Clinical Modification* [ICD-10-CM] codes Z55-Z65).

^e ICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^f Other variables that were propensity score matched between cohorts but were not shown include chemotherapy-specific medications (eg, taxanes, platinum compounds, anthracyclines and related substances, and cyclophosphamide).

^g Mastectomy, partial (eg, lumpectomy, tylectomy, quadrantectomy, segmentectomy).

nonuse (Figure 1E). In unadjusted models, GLP-1 RA use was associated with a statistically significantly lower hazard of RFS compared with insulin or metformin (HR, 0.33; 95% CI, 0.21-0.50; $P < .001$; Figure 3). In adjusted Cox proportional hazards regression model, GLP-1 RA use remained associated with a lower hazard of recurrence (HR, 0.44; 95% CI, 0.36-0.54; $P < .001$) (eTable 10 in Supplement 1).

Comparison of GLP-1 RA Use vs SGLT2 Inhibitors Among Patients With BC and T2D

Prior to PSM, the GLP-1 RA group included 11 689 patients, and the SGLT2 inhibitor group included 6096 patients. After PSM, groups were balanced with 4052 patients in each group (Table 1). Survival probabilities at 5 and 10 years were 86.3% and 75.5%, respectively, for GLP-1 RA use and 88.9% and 73.4%, respectively, for SGLT2 inhibitor use (Figure 1C). Comparison of GLP-1 RA use vs SGLT2 inhibitor use showed no significant differences in unadjusted risk for all-cause mortality (HR, 0.97; 95% CI, 0.82-1.14; $P = .72$) (Figure 2). In adjusted models, GLP-1 RA use was associated with a lower hazard of all-cause mortality (HR, 0.77; 95% CI, 0.68-0.88; $P < .001$) (eTable 9 in Supplement 1).

For RFS, prior to PSM, the GLP-1 RA group included 11 689 patients, and the SGLT2 inhibitor group included 6096 patients. After PSM, the 2 groups were balanced with 4000 patients in each group (eTable 7 in Supplement 1). Survival probabilities at 5 and 10 years for GLP-1 RA use were 95.0% and 90.9%, respectively, and for SGLT2 inhibitor use were 93.8% and 89.9%, respectively (Figure 1F). In unadjusted models, GLP-1 RA use was not associated with a statistically significant difference in recurrence compared with SGLT2 inhibitor use (HR, 0.91; 95% CI, 0.71-1.18; $P = .46$) (Figure 3). In adjusted models, GLP-1 RA use was associated with a significantly lower hazard of recurrence compared with SGLT2 inhibitor use (HR, 0.84; 95% CI, 0.73-0.97; $P = .02$) (eTable 10 in Supplement 1).

Subgroup and Sensitivity Analyses

Subgroup analyses of postmenopausal patients and landmark analyses (6 and 12 months) for all-cause mortality and RFS confirmed the robustness of the primary outcomes in the GLP-1 RA use vs nonuse and GLP-1 RA use vs insulin or metformin use (Table 2). No significant associations were observed when comparing GLP-1 RA use with SGLT2 inhibitor use among postmenopausal patients, and the protective association was not sustained in the 12-month landmark analyses for all-cause mortality (HR, 1.04; 95% CI, 0.80-1.35) or RFS (HR, 0.96; 95% CI, 0.73-1.30) (Table 2).

Discussion

The long-term use of GLP-1 RAs among patients with BC and obesity (without T2D) and patients with T2D at 5- and 10-year follow-up was associated with improved all-cause mortality and RFS vs insulin or metformin, but not with SGLT2 inhibitors. These associations (HRs, 0.09-0.44) suggest a meaningful difference in survival and recurrence outcomes across comparison groups. Subgroup and landmark analyses confirmed the robustness of these findings, supporting the potential for GLP-1 RAs to improve survival and reduce risk of recurrence among patients with BC and obesity or T2D.

Patients with BC and metabolic comorbidities face increased risk of cancer recurrence and reduced survival.¹⁻⁵ Although studies of patients with BC have shown the benefit of GLP-1 RAs on weight loss^{18-23,40} and related comorbidities²⁴ (eg, cardiovascular events independent of weight loss),^{25,26} few have focused on cancer outcomes. Our results from real-world data suggest that GLP-1 RA use potentially improves all-cause mortality and reduces the risk of RFS among patients with BC

Table 2. Landmark and Subgroup Analyses of All-Cause Mortality and Recurrence-Free Survival Over a 10-Year Follow-Up Period

Outcome	No. of deaths/No. of patients		HR (95% CI)
	GLP-1 RA	Comparison group	
All-cause mortality			
Landmark 6 mo ^a			
GLP-1 RA use vs nonuse	27/1476	42/1479	0.68 (0.42-1.11)
GLP-1 RA vs insulin or metformin	17/2120	121/2123	0.14 (0.08-0.23)
GLP-1 RA vs SGLT2 inhibitor	126/3383	112/3296	1.09 (0.84-1.41)
Landmark 12 mo ^a			
GLP-1 RA use vs nonuse	22/1363	37/1365	0.63 (0.37-1.07)
GLP-1 RA vs insulin or metformin	16/1830	107/1827	0.14 (0.08-0.24)
GLP-1 RA vs SGLT2 inhibitor	113/2990	108/2958	1.04 (0.80-1.35)
Postmenopausal			
GLP-1 RA use vs nonuse	22/1309	42/1312	0.55 (0.33-0.93)
GLP-1 RA vs insulin or metformin	16/2154	154/2154	0.10 (0.06-0.16)
GLP-1 RA vs SGLT2 inhibitor	117/3538	111/3461	1.01 (0.78-1.30)
Recurrence-free survival			
Landmark 6 mo			
GLP-1 RA use vs nonuse	79/1442	117/1442	0.55 (0.38-0.82)
GLP-1 RA vs insulin or metformin	30/1923	73/1844	0.38 (0.25-0.58)
GLP-1 RA vs SGLT2 inhibitor	124/3384	118/3320	1.02 (0.80-1.31)
Landmark 12 mo			
GLP-1 RA use vs nonuse	35/1223	69/1238	0.50 (0.33-0.75)
GLP-1 RA vs insulin or metformin	29/2109	78/1981	0.33 (0.21-0.50)
GLP-1 RA vs SGLT2 inhibitor	105/3021	108/2975	0.96 (0.73-1.30)
Postmenopausal			
GLP-1 RA use vs nonuse	37/1181	76/1207	0.48 (0.32-0.71)
GLP-1 RA vs insulin or metformin	27/1957	80/1850	0.30 (0.19-0.46)
GLP-1 RA vs SGLT2 inhibitor	106/3552	115/3495	0.91 (0.70-1.19)

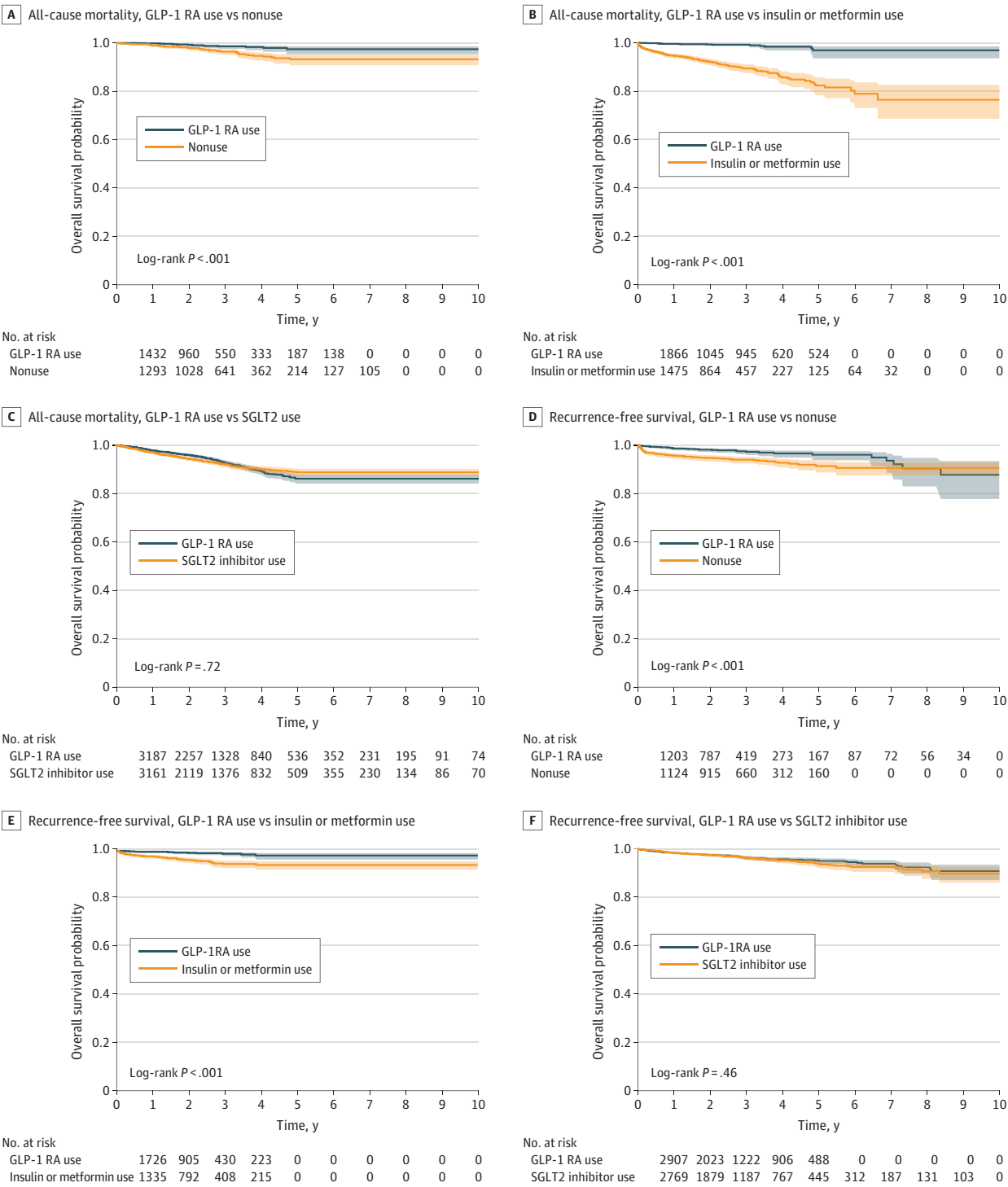
Abbreviations: GLP-1 RA, glucagon-like peptide-1 receptor agonist; HR, hazard ratio; SGLT2, sodium-glucose cotransporter 2.

^a Landmark analysis defined the cohorts at the specified time points (6 and 12 months) to mitigate immortal time bias.

and obesity and related metabolic conditions, contributing to a growing body of literature related to GLP-1 RA use in oncology settings.⁴¹⁻⁴³

The findings align with emerging mechanistic and preclinical research. For instance, a study evaluating the prognostic importance of GLP-1 receptor expression across cancer types found that in

Figure 1. Survival Plots Showing 10-Year All-Cause Mortality and Recurrence-Free Survival by Glucagon-Like Peptide-1 Receptor Agonist (GLP-1 RA) Use Compared With Other Antidiabetic Medications



The shaded areas indicate 95% CIs. SGLT2 indicates sodium glucose cotransporter 2.

certain cancers, including BC, tumors with higher GLP-1 receptor expression were associated with improved overall survival compared with tumors with lower GLP-1 receptor expression. These findings suggest a potential mechanistic role of GLP-1 signaling for certain cancers.⁴⁴ Emerging preclinical mouse models of BC and endometrial cancer are also finding evidence for antitumorigenic effects of GLP-1 RAs (tirzepatide) via metabolic and immune pathways^{27,45} Mechanistic studies are needed to understand the causal pathways of GLP-1 RAs in BC outcomes.

Among patients with BC, obesity at diagnosis and weight gain during therapy worsen outcomes, potentially through tumor growth increasing recurrence risk. Patients with BC taking GLP-1 RAs may experience weight loss and/or improved cardiovascular health. The clinical utility of GLP-1 RAs in cancer care is evolving.⁴⁶ Certain antidiabetic and antiobesity medications may have differential effects among patients with BC. For instance, SGLT2 inhibitors can improve cardiometabolic and heart failure outcomes, while GLP-1 RAs provide greater glycemic control and weight loss than other antidiabetic medications, including dipeptidyl peptidase-4 inhibitors or those with longer clinical histories, such as metformin.⁴⁷ Preliminary evidence suggests reduced weight-loss efficacy of GLP-1 RAs among patients with BC undergoing endocrine therapy.⁴³ Clinical use of GLP-1 RAs among patients receiving endocrine therapy warrants further investigation.

GLP-1 RAs were approved for T2D treatment in 2005 and for weight management among adults with a BMI of 30 or more (or ≥ 27 with at least 1 weight-related comorbidity) in 2021.^{15,17} Since this time, the use of GLP-1 RAs has markedly increased, with approximately 1 in 8 US adults reporting ever using a GLP-1 RA, including 40% of those with diabetes and 25% of those with heart disease.⁴⁸ Some patients may have received GLP-1 RAs off label prior to formal FDA approval for obesity; such use would likely represent a minority of patients and would be unlikely to meaningfully affect exposure duration patterns in our results. Replication of these findings is warranted with extended follow-up in the TriNetX database. Our results may underestimate the 5- and 10-year survival effects, especially among those using GLP-1 RAs for weight management or cardiovascular disease rather than for T2D. Newer agents (tirzepatide and oral GLP-1 RAs) are also becoming available, and it is unclear how the changing landscape might affect cancer outcomes. Furthermore, as data accumulate, it may also be possible to evaluate the timing of GLP-1 RA treatment (prediagnosis use or for the duration of

Figure 2. Forest Plot Showing Hazard Ratios (HRs) for All-Cause Mortality by Glucagon-Like Peptide-1 Receptor Agonist (GLP-1 RA) Use Compared With Other Antidiabetic Medications

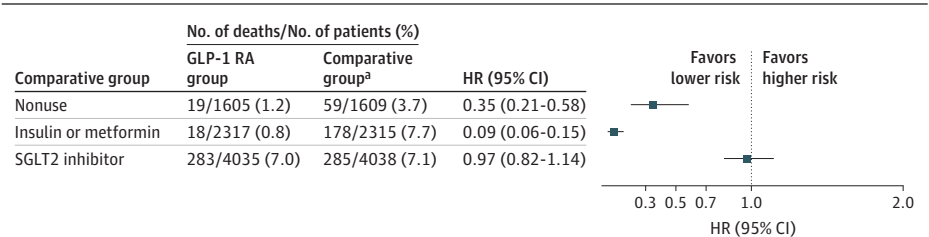
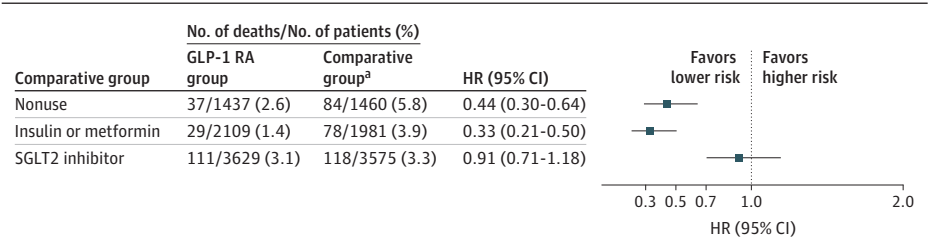


Figure 3. Forest Plot Showing Hazard Ratios (HRs) for Recurrence-Free Survival by Glucagon-Like Peptide-1 Receptor Agonist (GLP-1RA) Use Compared With Other Antidiabetic Medications



treatment) on cancer outcomes. Our study underscores the potential of GLP-1 RAs as an adjunct strategy for improving cancer-related outcomes among patients with BC, although clinical trials are needed to inform effective therapeutic approaches and clinical decision-making.

Strengths and Limitations

This study has some strengths, including the examination of long-term survival at 10 years, the 1:1 PSM model, and the inclusion of other antidiabetic medications (insulin, metformin, and SGLT2 inhibitors). We also included subgroup analyses with a higher-risk group (postmenopausal women) and landmark analyses to mitigate immortal-time bias.

This study also has several limitations. Its retrospective design and reliance on a US-based database limit generalizability and introduce potential bias and misdiagnoses. Although we controlled for specific variables, causality cannot be inferred. TriNetX patient-level data should be validated against other EHR databases. Approximation of the 25th and 75th percentiles assumes a symmetric distribution and may not fully reflect skewness in the underlying data. Diagnoses were based on diagnostic codes in EHRs; thus, the accuracy of cancer characteristics and causes of death is unclear. There is no standard definition of disease progression (eg, recurrence, metastasis); however, we used previously published EHR studies to guide and define the coding of cancer recurrence.^{35,36} Another limitation is that although this database is larger, PSM reduces the number of patients available for outcomes and certain comparison groups. In addition, the substantial administrative censoring around 5 years reduced the number of patients remaining at risk at later time points, which limits the precision of the 10-year Kaplan-Meier survival estimates and HRs.⁴⁹ GLP-1 RAs were grouped without individual assessment, limiting our understanding of mechanistic variations. Although 2 or more prescriptions were used as an initiation proxy, it remains unclear whether the observed long-term outcomes reflect initial metabolic changes or sustained therapy. These findings are real-world associations with treatment initiation rather than the biological effects of continuous exposure. Because EHRs often lack patient medication adherence data, we used an intention-to-treat approach based on prescriptions. Patient-level weight change data were unavailable; therefore, the mechanistic drivers of the observed survival effects (eg, weight loss) remain unclear.²⁷ The lower-than-expected proportion of estrogen receptor-positive or estrogen receptor-negative tumors likely reflects incomplete capture of oncologic characteristics in structured EHR fields within TriNetX. This limitation is inherent to real-world EHR data and may affect other tumor and treatment characteristics.

Conclusions

In this cohort study of patients with BC, our results suggest promising associations between GLP-1 RA use and improved outcomes after a BC diagnosis. However, considerable gaps remain in our understanding of the role of GLP-1 RAs in oncology, including the mechanistic pathways associated with BC outcomes. Prospective clinical studies are warranted to understand the potential risks and the optimal timing and tailoring of GLP-1 RA treatment for patients with BC.

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Author Contributions: Dr Tatum had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Administrative, technical, or material support: Stevenson, Williford, Cassel, Zhao, Diala, Hundley.

Supervision: Tatum, Dahman, Williford, Fuemmeler.

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REFERENCES

1. Barone BB, Yeh HC, Snyder CF, et al. Long-term all-cause mortality in cancer patients with preexisting diabetes mellitus: a systematic review and meta-analysis. *JAMA*. 2008;300(23):2754-2764. doi:10.1001/jama.2008.824
2. Peairs KS, Barone BB, Snyder CF, et al. Diabetes mellitus and breast cancer outcomes: a systematic review and meta-analysis. *J Clin Oncol*. 2011;29(1):40-46. doi:10.1200/JCO.2009.27.3011
3. Zhao XB, Ren GS. Diabetes mellitus and prognosis in women with breast cancer: a systematic review and meta-analysis. *Medicine (Baltimore)*. 2016;95(49):e5602. doi:10.1097/MD.0000000000005602
4. Li CI, Daling JR, Tang MTC, Malone KE. Relationship between diabetes and risk of second primary contralateral breast cancer. *Breast Cancer Res Treat*. 2011;125(2):545-551. doi:10.1007/s10549-010-1035-4
5. Kaplan MA, Pekkola Z, Kucukoner M, et al. Type 2 diabetes mellitus and prognosis in early stage breast cancer women. *Med Oncol*. 2012;29(3):1576-1580. doi:10.1007/s12032-011-0109-4
6. Petrelli F, Cortellini A, Indini A, et al. Association of obesity with survival outcomes in patients with cancer: a systematic review and meta-analysis. *JAMA Netw Open*. 2021;4(3):e213520. doi:10.1001/jamanetworkopen.2021.3520

7. Chan DSM, Vieira AR, Aune D, et al. Body mass index and survival in women with breast cancer—systematic literature review and meta-analysis of 82 follow-up studies. *Ann Oncol*. 2014;25(10):1901-1914. doi:10.1093/annonc/mdl042
8. Shao S, Gill AA, Zahm SH, et al. Diabetes and overall survival among breast cancer patients in the US military health system. *Cancer Epidemiol Biomarkers Prev*. 2018;27(1):50-57. doi:10.1158/1055-9965.EPI-17-0439
9. Dibaba DT, Ogunsina K, Braithwaite D, Akinyemiju T. Metabolic syndrome and risk of breast cancer mortality by menopause, obesity, and subtype. *Breast Cancer Res Treat*. 2019;174(1):209-218. doi:10.1007/s10549-018-5056-8
10. Arnold M, Pandeya N, Byrnes G, et al. Global burden of cancer attributable to high body-mass index in 2012: a population-based study. *Lancet Oncol*. 2015;16(1):36-46. doi:10.1016/S1470-2045(14)71123-4
11. Lee E, Kawaguchi ES, Zhang J, et al. Bariatric surgery in patients with breast and endometrial cancer in California: population-based prevalence and survival. *Surg Obes Relat Dis*. 2022;18(1):42-52. doi:10.1016/j.soard.2021.09.017
12. Wooster M, Chen L, Accordini MK, Sathe C, Wright JD, Hershan DL. Utilization of weight management treatment and subsequent cardiovascular events among patients with breast cancer. *Breast Cancer Res Treat*. 2025;212(2):217-223. doi:10.1007/s10549-025-07714-6
13. Goodwin PJ, Segal RJ, Vallis M, et al. The LISA randomized trial of a weight loss intervention in postmenopausal breast cancer. *NPJ Breast Cancer*. 2020;6:6. doi:10.1038/s41523-020-0149-z
14. Radwan RM, Lu Y, Dai H, et al. GLP-1 RA use and survival among older adults with cancer and type 2 diabetes. *JAMA Netw Open*. 2025;8(7):e2521887. doi:10.1001/jamanetworkopen.2025.21887
15. Update on FDA's ongoing evaluation of reports of suicidal thoughts or actions in patients taking a certain type of medicines approved for type 2 diabetes and obesity. US Food and Drug Administration. 2024. Accessed January 10, 2025. <https://www.fda.gov/drugs/drug-safety-and-availability/update-fdas-ongoing-evaluation-reports-suicidal-thoughts-or-actions-patients-taking-certain-type>
16. Drucker DJ. GLP-1 physiology informs the pharmacotherapy of obesity. *Mol Metab*. 2022;57:101351. doi:10.1016/j.molmet.2021.101351
17. FDA approves new drug treatment for chronic weight management, first since 2014. US Food and Drug Administration. June 4, 2021. Accessed March 31, 2026. <https://www.prnewswire.com/news-releases/fda-approves-new-drug-treatment-for-chronic-weight-management-first-since-2014-301306223.html>
18. Jensterle M, Rizzo M, Haluzik M, Janež A. Efficacy of GLP-1 RA approved for weight management in patients with or without diabetes: a narrative review. *Adv Ther*. 2022;39(6):2452-2467. doi:10.1007/s12325-022-02153-x
19. Wilding JPH, Batterham RL, Calanna S, et al; STEP 1 Study Group. Once-weekly semaglutide in adults with overweight or obesity. *N Engl J Med*. 2021;384(11):989-1002. doi:10.1056/NEJMoa2032183
20. Astrup A, Rössner S, Van Gaal L, et al; NN8022-1807 Study Group. Effects of liraglutide in the treatment of obesity: a randomised, double-blind, placebo-controlled study. *Lancet*. 2009;374(9701):1606-1616. doi:10.1016/S0140-6736(09)61375-1
21. Wadden TA, Tronieri JS, Sugimoto D, et al. Liraglutide 3.0 mg and intensive behavioral therapy (IBT) for obesity in primary care: the SCALE IBT randomized controlled trial. *Obesity (Silver Spring)*. 2020;28(3):529-536. doi:10.1002/oby.22726
22. Wang JY, Wang QW, Yang XY, et al. GLP-1 receptor agonists for the treatment of obesity: role as a promising approach. *Front Endocrinol (Lausanne)*. 2023;14:1085799. doi:10.3389/fendo.2023.1085799
23. Jastreboff AM, le Roux CW, Stefanski A, et al; SURMOUNT-1 Investigators. Tirzepatide for obesity treatment and diabetes prevention. *N Engl J Med*. 2025;392(10):958-971. doi:10.1056/NEJMoa2410819
24. Popoviciu MS, Păduraru L, Yahya G, Metwally K, Cavalu S. Emerging role of GLP-1 agonists in obesity: a comprehensive review of randomised controlled trials. *Int J Mol Sci*. 2023;24(13):10449. doi:10.3390/ijms241310449
25. Marx N, Husain M, Lehrke M, Verma S, Sattar N. GLP-1 receptor agonists for the reduction of atherosclerotic cardiovascular risk in patients with type 2 diabetes. *Circulation*. 2022;146(24):1882-1894. doi:10.1161/CIRCULATIONAHA.122.059595
26. Zelniker TA, Wiviott SD, Raz I, et al. Comparison of the effects of glucagon-like peptide receptor agonists and sodium-glucose cotransporter 2 inhibitors for prevention of major adverse cardiovascular and renal outcomes in type 2 diabetes mellitus. *Circulation*. 2019;139(17):2022-2031. doi:10.1161/CIRCULATIONAHA.118.038868
27. Glenn EM, Ho AN, Kiesel VA, et al. Tirzepatide attenuates mammary tumor progression in diet-induced obese mice. *bioRxiv*. Preprint posted online January 23, 2024. doi:10.1101/2024.01.20.576484

28. Ligumsky H, Wolf I, Israeli S, et al. The peptide-hormone glucagon-like peptide-1 activates cAMP and inhibits growth of breast cancer cells. *Breast Cancer Res Treat*. 2012;132(2):449-461. doi:10.1007/s10549-011-1585-0
29. Malin B. Summary of the assessment of TriNetX data privacy principles through an empirical analysis. TriNetX. 2020. Accessed November 10, 2025. <https://trinetx.com/wp-content/uploads/2021/12/TriNetX-Empirical-Summary-by-Brad-Malin-2020.pdf>
30. TriNetX. September 8, 2021. Accessed February 1, 2025. <https://trinetx.com/>
31. Palchuk MB, London JW, Perez-Rey D, et al. A global federated real-world data and analytics platform for research. *JAMIA Open*. 2023;6(2):o0ad035. doi:10.1093/jamiaopen/o0ad035
32. Updates to race/ethnicity standards for our nation. U.S. Census Bureau. 2024. Accessed November 10, 2025. <https://www.census.gov/about/our-research/race-ethnicity/standards-updates.html>
33. Wang L, Xu R, Kaelber DC, Berger NA. Glucagon-like peptide 1 receptor agonists and 13 obesity-associated cancers in patients with type 2 diabetes. *JAMA Netw Open*. 2024;7(7):e2421305. doi:10.1001/jamanetworkopen.2024.21305
34. Wang L, Wang W, Kaelber DC, Xu R, Berger NA. GLP-1 receptor agonists and colorectal cancer risk in drug-naive patients with type 2 diabetes, with and without overweight/obesity. *JAMA Oncol*. 2024;10(2):256-258. doi:10.1001/jamaoncol.2023.5573
35. Lord SJ, Kiely BE, Pearson SA, et al. Metastatic breast cancer incidence, site and survival in Australia, 2001-2016: a population-based health record linkage study protocol. *BMJ Open*. 2019;9(2):e026414. doi:10.1136/bmjopen-2018-026414
36. Holloway CMB, Shabestari O, Eberg M, et al. Identifying breast cancer recurrence in administrative data: algorithm development and validation. *Curr Oncol*. 2022;29(8):5338-5367. doi:10.3390/curroncol29080424
37. Charlson ME, Carrozzino D, Guidi J, Patierno C. Charlson Comorbidity Index: a critical review of clinimetric properties. *Psychother Psychosom*. 2022;91(1):8-35. doi:10.1159/000521288
38. Sharma N, Narayan S, Sharma R, Kapoor A, Kumar N, Nirban R. Association of comorbidities with breast cancer: an observational study. *Trop J Med Res*. 2015;19:168-171. doi:10.4103/1110-0388.185449
39. Fleming ST, Rastogi A, Dmitrienko A, Johnson KD. A comprehensive prognostic index to predict survival based on multiple comorbidities: a focus on breast cancer. *Med Care*. 1999;37(6):601-614. doi:10.1097/00005650-199906000-00009
40. Shen S, Liu B, Fanti C, et al. GLP-1 receptor agonist use and weight change in patients with breast cancer. *Oncology (Williston Park)*. 2025;39(7):294-296. doi:10.46883/2025.25921046
41. Shen S, Lui B, Fanti C, Bromberg M, Chen Y, Lyengar N. Glucagon-like peptide-1 (GLP-1) agonist use and weight change among patients with breast cancer. *J Clin Oncol*. 2024;42(16):10607. doi:10.1200/JCO.2024.42.16_suppl.10607
42. Fischbach NA, Zhou B, Deng Y, Parsons K, Shelton A, Lustberg MB. Impact of semaglutide and tirzepatide administration on weight in women with stage I-III breast cancer. *J Clin Oncol*. 2024;42(16)(suppl):e24140. doi:10.1200/JCO.2024.42.16_suppl.e24140
43. Urueta Portillo D, Mazo-Canola M, Alhaj S. Evaluating the impact of GLP-1 receptor agonists on weight management in patients with breast cancer undergoing endocrine therapy: a comparative analysis. *J Clin Oncol*. 2024;42(16)(suppl):e13063. doi:10.1200/JCO.2024.42.16_suppl.e13063
44. Ungvari Z, Bartha Á, Ungvari A, Fekete M, Bianchini G, Györffy B. Prognostic impact of glucagon-like peptide-1 receptor (GLP1R) expression on cancer survival and its implications for GLP-1R agonist therapy: an integrative analysis across multiple tumor types. *Geroscience*. 2025;47(3):4413-4427. doi:10.1007/s11357-024-01494-5
45. Kong W, Deng B, Shen X, et al. Tirzepatide as an innovative treatment strategy in a pre-clinical model of obesity-driven endometrial cancer. *Gynecol Oncol*. 2024;191:116-123. doi:10.1016/j.ygyno.2024.10.004
46. Parsons K, Montalvo M, Fischbach N, Taylor M, Alfaro S, Lustberg M. The impact and safety of glp-1 agents and breast cancer. *Cancer Med*. 2025;14(12):e70932. doi:10.1002/cam4.70932
47. Brunton S. GLP-1 receptor agonists vs DPP-4 inhibitors for type 2 diabetes: is one approach more successful or preferable than the other? *Int J Clin Pract*. 2014;68(5):557-567. doi:10.1111/ijcp.12361
48. Montero A, Sparks G, Presiado M, Hamel L. KFF Health tracking poll may 2024: the public's use and views of glp-1 drugs. May 10, 2024. Accessed May 26, 2026. <https://www.kff.org/health-costs/poll-finding/kff-health-tracking-poll-may-2024-the-publics-use-and-views-of-glp-1-drugs/>
49. Rich JT, Neely JG, Paniello RC, Voelker CCJ, Nussenbaum B, Wang EW. A practical guide to understanding Kaplan-Meier curves. *Otolaryngol Head Neck Surg*. 2010;143(3):331-336. doi:10.1016/j.otohns.2010.05.007

SUPPLEMENT 1.

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SUPPLEMENT 2.

Data Sharing Statement